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IAPP GLOBAL PRIVACY SUMMIT 2022

Evaluating Algorithms: Incorporating Privacy and Ethics into AI/ML Projects

WELCOME AND INTRODUCTIONS

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Global regulation of automated decision-making

- GDPR Art. 22: "The data subject shall have the right not to be subject to a decision <u>based solely on automated processing</u>, including profiling, which produces <u>legal effects</u> concerning him or her or <u>similarly significantly affects</u> him or her."
- Similar rules in Brazil (LGPD), South Africa (POPIA), and China (PIPL)

ADM laws: Coming to a state near you

California (CPRA)

- CPRA rulemaking to produce regulations "governing **access** and **opt-out rights** with respect to business' use of automated decision-making technology, including **profiling** and requiring businesses' response to access requests to include **meaningful information** about the logic involved in such decision-making processes, as well as a **description of the likely outcome** of the process with respect to the consumer"

- "Profiling" means "any form of automated processing of personal information [as defined in rulemaking] to **evaluate** certain personal aspects relating to a natural person, and in particular to **analyze or predict** aspects concerning that natural person's performance at work, economic situation, health, personal preferences, interests, reliability, behavior, location, or movements"

Current US regulation of algorithms

- **FTC guidance:**
  - Sale or use of racially biased algorithms may be “unfair”
  - Recent guidance focuses on transparency, unexaggerated claims, and vetting algorithms before launch
  - Watch for claims of data deletion, ability to opt in/out

- **Civil rights/anti-discrimination**
  - Fair Credit Reporting Act (FCRA)
  - Equal Credit Opportunities Act (ECOA)
  - AI bills/resolutions introduced in multiple states

Pending EU AI Regulation

• Manipulation leading to psychological/physical harms or on age/disability vulnerability • Decontextualized or unjustifiably detrimental social scoring system • Indiscriminate facial recognition by police	Prohibition
• Credit scoring; assessment of workers, students • AI used in critical infrastructure; by judges, public administration, police, border control • Products subject to safety regulations under EU law	Design obligations (human oversight, data management plan, risk assessment, etc.)
• Deep fake • Emotion recognition • AI interacting with humans	Transparency obligations
• Residual	Voluntary codes of conduct

Common threads across laws

- Privacy impact assessments and consumer rights to opt-out or limit decisions, especially for higher-risk applications
- Application may vary depending on harm/impact
  - Rules applying to “decisions that produce legal or similarly significant effects”
  - Differing level of diligence depending on potential harm
- Discriminatory or disparate impact likely to trigger additional regulatory scrutiny
- Laws may require transparency/consent, even for “minimal risk” applications

So where do we start?

GDPR privacy principles (Art. 5) / fair information practices model

Processing must comply with the following principles:

- Lawfulness, fairness, and transparency
- Purpose limitation
- Data minimization
- Accuracy
- Storage limitation
- Integrity and confidentiality
- Accountability

So where do we start?

GDPR privacy principles (Art. 5) / fair information practices model

Processing must comply with the following principles:

- Lawfulness, **FAIRNESS**, and **TRANSPARENCY**
- Purpose limitation
- Data minimization
- Accuracy
- Storage limitation
- Integrity and confidentiality
- **ACCOUNTABILITY**
- **EXPLAINABILITY**

Taken together, these concepts raise the greater concerns with machine learning models

Fairness

Algorithmic decisions should not create **discriminatory or unjust impacts** when comparing across different demographics or affected communities and individuals

Explainability

- Determining how the algorithm works
- But this can be a **“black box problem”**

Risk discussions must occur during the development & design of an AI/ML model

Transparency

The personal data collected, the model's function and process, and the role of the model's output in final decision-making are **understandable to the user or individual** impacted by the decision

Accountability

Algorithmically informed decisions have the potential for significant social impact, and should be **designed and implemented in publicly accountable ways**, such as by reporting, explaining, and justifying specific decisions as well as mitigating negative impacts and potential harms

Fairness

Consider potential risks to fairness throughout the product lifecycle:

1. Data acquisition strategy
   a) How to make sure that data sets are sufficiently representative?
   b) Can/should we use company data, public data, synthetic data?
      1) Open data
      2) Alternative data

Fairness

Consider potential risks to fairness throughout the product lifecycle:

2. Consider developing an AI/ML ops platform
   a) Standard data sets
   b) Team can plug into the sandbox
   c) Avoids disparate reviews, forum shopping

Fairness

Consider potential risks to fairness throughout the product lifecycle:

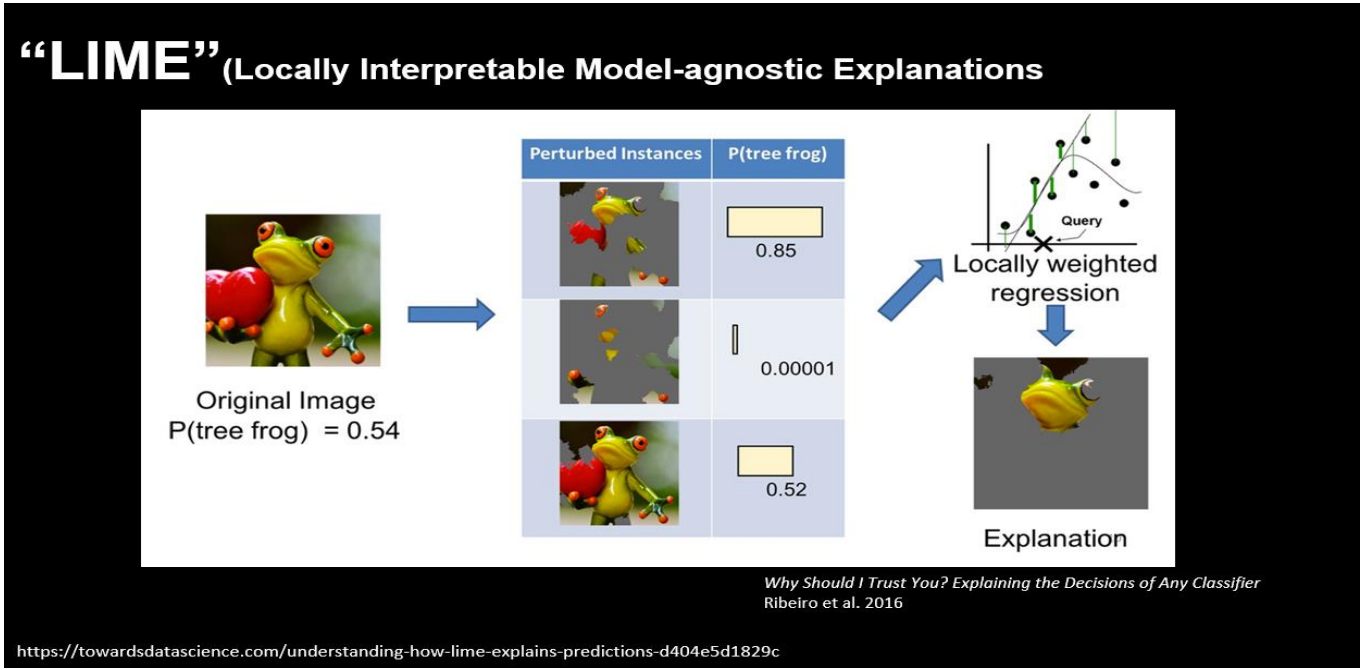
3. Determine triggers for review, escalation
   a) Should this be a purely legal decision, or should there be a cross-functional committee?
   b) How much can be self-serve / programmatized?

Fairness – Example of AI Principles

1. Be socially beneficial
2. Avoid creating or reinforcing unfair bias
3. Be built and tested for safety
4. Be accountable to people
5. Incorporate privacy design principles
6. Uphold high standards of scientific excellence
7. Be made available for uses that accord with these principles

[Artificial Intelligence at Google: Our Principles, https://ai.google/principles/]

Explainability in practice



[Tomsett, R. (2020). Explainable AI (XAI). In *Digital Data Flows Masterclass: Understanding Machine Learning*, Future of Privacy Forum, CPDP 2020 side event]

Explainability in practice

(a) Husky classified as wolf (b) Explanation

Why Should I Trust You? Explaining the Decisions of Any Classifier
Ribeiro et al. 2016

[Tomsett, R. (2020). Explainable AI (XAI). In *Digital Data Flows Masterclass: Understanding Machine Learning*, Future of Privacy Forum, CPDP 2020 side event]

Tools for explainability

Triggers	User/Learner's Goal	
	Need to Understand	Need to Accomplish
How do I use it?	Use	Primary task goals
How does it work?	Mechanism, Understandability	Feeling of Satisfaction
What did it just do?	Mechanism, Understandability	Feeling of Sufficiency
What does it achieve?	Function	Usefulness
What will it do next?	Observability	Trust, Reliance
How much effort will this take?	Usability	Primary Task Goals
What do I do if it gets it wrong?	Anticipation	Control
How do I avoid the failure modes?	Surprise	Adaptation, Work-around
What would it have done if x was different?	Mechanism, Understandability	Curiosity, Trust, Reliance
Why didn't it do z?	Anticipation, Surprise	Curiosity, Trust, Reliance

Mueller, S. T., Hoffman, R. R., Clancey, W., Emrey, A., & Klein, G. (2019). Explanation in human-AI systems: A literature meta-review, synopsis of key ideas and publications, and bibliography for explainable AI, p. 90.

Tools for explainability (cont'd)

Format (How it is expressed)	Reference (What it is about)
• Visualizations (e.g., Heat maps) • Text (Statements, Narratives or Stories, Answers to queries, Human-machine dialogs) • Formal Expressions (Logical expressions, Matrices) • Conceptual Process Models (Diagrams) • Graphs, Networks • Tables • Abstractions, generalizations • Timelines • Hierarchies (Trees)	• Examples (includes misclassifications, counter-examples, outliers, clear cases, close competitors) • Patterns, Classes, Ontologies • Features, Weights, Probabilities, Ranks, Parameters • Decisions, Strategies, Goals • Algorithms, Computational Processes, Proofs • Incidents, Events (includes self-explanations or stories) • Cause-effect relations

Mueller, S. T., Hoffman, R. R., Clancey, W., Emrey, A., & Klein, G. (2019). Explanation in human-AI systems: A literature meta-review, synopsis of key ideas and publications, and bibliography for explainable AI, p. 84.

Tools for explainability (cont'd)

Fairness metrics	Bias preserving?
1. Group fairness, Statistical (demographic) parity[i]	✗
2. Conditional statistical (demographic) parity, Conditional independence[ii]	✗
3. Predictive parity, outcome test[iii]	✓
4. False positive error rate balance[iv]	✓
5. False negative error rate balance,[v] Equal opportunity[vi]	✓
6. Equalized odds[vii]	✓
7. Conditional use accuracy equality[viii]	✓
8. Overall accuracy equality[ix]	✓
9. Treatment equality[x]	✓
10. Test-fairness or calibration[xi]	✓
11. Well-calibration[xii]	✓
12. Balance for positive class[xiii]	✓
13. Balance for negative class[xiv]	✓
14. Causal discrimination[xv] (direct discrimination)	*
15. Fairness through unawareness[xvi]	*
16. Fairness through awareness[xvii]	✗
17. Counterfactual fairness[xviii]	✗
18. No unresolved discrimination[xix] (causal variant of 2)	✗
19. No proxy discrimination[xx]	✗
20. Path based causal reasoning[xxi]	✗

Wachter S., Mittelstat B. & Rusell C. (2021). Bias Preservation in Machine Learning: The Legality of Fairness Metrics Under EU Non-Discrimination Law. In: West Virginia Law Review, Forthcoming.

Tools for explainability (cont'd)

Explainability

Guiding Questions

- Who are your end-users and stakeholders?
- How much of your system / algorithm can you explain to your users and stakeholders?
- How much of the data sources can you disclose?

Initial Steps to Take

- Have a plan for how decisions will be explained to users and subjects of those decisions. In some cases it may be appropriate to develop an automated explanation for each decision.
- Allow data subjects visibility into the data you store about them and access to a process in order to change it.
- If you are using a machine-learning model:
  - consider whether a directly interpretable or explainable model can be used.
  - describe the training data including how, when, and why it was collected and sampled.
  - describe how and when test data about an individual that is used to make a decision is collected or inferred.
- Disclose the sources of any data used and as much as possible about the specific attributes of the data. Explain how the data was cleaned or otherwise transformed.

https://www.fatml.org/resources/principles-for-accountable-algorithms

Transparency

Accountability

Counseling Considerations

- Don't get tunnel vision or distracted by a single issue
- Consider an issue-spotting checklist, *e.g.*
  - Fairness (bias, discrimination, potential harms)
  - Explainability
  - Transparency
  - Accountability
  - Privacy & data governance
  - Automated decision making (profiling)
  - Personalization
  - Accessibility
  - Copyright
  - Competition
  - Product liability
  - Law enforcement access
  - Third party concerns

Questions?

RESOURCE LIST

- GDPR Article 5 – Principles relating to processing of data
- Future of Privacy Forum (2018). The Expert's Guide To Artificial Intelligence and Machine Learning
- Artificial Intelligence at Google: Our Principles
- Tomsett, R. (2020). Explainable AI (XAI). In *Digital Data Flows Masterclass: Understanding Machine Learning*, Future of Privacy Forum, CPDP 2020 side event
- Mueller, S. T., Hoffman, R. R., Clancey, W., Emrey, A., & Klein, G. (2019). Explanation in human-AI systems: A literature meta-review, synopsis of key ideas and publications, and bibliography for explainable AI
- Wachter S., Mittelstat B. & Rusell C. (2021). Bias Preservation in Machine Learning: The Legality of Fairness Metrics Under EU Non-Discrimination Law. In: West Virginia Law Review, Forthcoming
- Principles for Accountable Algorithms and a Social Impact Statement for Algorithms

How Did Things Go?
(We Really Want To Know)

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